KIET GROUP OF INSTITUTIONS, GHAZIABAD

CSE (AI/AIML) Department | Project (BCS-753) | 7th Semester

Multilingual Mental-Health Risk and Sentiment Analyzer

Screening sentiment, emotion and stress-related language in English, Hindi and Hinglish, with a working web application and an IEEE-format research paper

Team members

Faisal Arif Ansari | **Paras Tiwari** | **Prateek Rawat** | **Lakshay Mishra**

Research and screening tool only. It does not diagnose any mental-health condition.

How this presentation maps to the evaluation

#	Assessment component	Where it is covered	Evidence shown
1	Technical approach finalized and validated	Slides 4, 6-8	Literature, alternatives compared, validation protocol
2	Implementation (minimum 70%)	Slides 10-15	Component status, live web app screenshots
3	Results analysis, discussion and validation	Slides 17-22	3-seed test results, 95% CIs, published baselines
4	Presentation	Whole deck	Live demo of the web app (slides 11-15)
5	Team work and individual contribution	Slide 24	Work packages and owners
6	Project management, cost and scalability	Slides 25-26	Timeline, measured compute, latency, scaling plan
7	Ethics, environment, sustainability, safety	Slide 28	Privacy design, safety layer, compute footprint
8	Research paper quality and plagiarism	Slide 29	7-page IEEE paper, 37 verified references
9	Submission readiness and communication	Slides 29-30	Paper, public code, reproducibility guide

Problem and scope

- Indian social-media text mixes English, Hindi in Devanagari and Hindi typed in Roman letters (Hinglish), often in one sentence.
- Most mental-health NLP tools are trained and tested on English only. Their behaviour on Hinglish is rarely measured.
- Goal: one service that reports sentiment, emotion and a stress-language signal, with language ID, calibrated probabilities and token-level explanations.
- Research question: what explains the English to Hinglish accuracy gap, and which change closes it?

Scope: screening and research only. The "risk" output means stress-related language in a post, as labelled in Dreaddit. It is not a clinical diagnosis.



3

tasks: sentiment, emotion, stress language



3

language and script tags: en-Latn, hi-Deva, hi-Latn



4

public datasets, no private data collected

What earlier research did, and the gap we target

Rubric 1

Prior work	What it showed	Gap addressed in this project
Turcan & McKeown, 2019 (Dreaddit)	Reddit stress dataset; BERT-base F1 0.8065 on the official test set	English only; we reuse its test set to compare directly
Khanuja et al., 2021 (MuRIL)	BERT pre-trained on Indian languages, including transliterated text	Not tested on Hinglish social-media sentiment with stress tasks
Nayak & Joshi, 2022 (HingBERT)	Hinglish pre-training: HingRoBERTa 66.43 vs XLM-R 61.57 F1 (GLUECoS)	We test whether the gain holds on new Hinglish data
Barbieri et al., 2022 (UMSAB)	Multilingual tweet sentiment; Hindi XLM-Twitter 56.39	Same Hindi test file reused for context
Muhammad et al., 2025 (BRIGHTER)	Multi-label emotion data for many languages, incl. Hindi	Used for the emotion task (English and Hindi)
Liu et al., 2019; Pruksachatkun et al., 2020	Multi-task training helps some task pairs and hurts others	Untested for unrelated mental-health datasets; we ablate it
Guo et al., 2017; Sundararajan et al., 2017	Temperature scaling; Integrated Gradients explanations	Explanation faithfulness on Hinglish is rarely measured

1

Technical approach

Data, architecture, alternatives compared and how each choice was validated

Datasets and split integrity

Rubric 1

Task	Dataset	Languages	Train / Val / Test
Sentiment (3 classes)	UMSAB tweets (English, "hindi") + Hinglish YouTube comments	en-Latn, hi-Latn	5,854 / 1,121 / 2,213
Emotion (6 labels)	BRIGHTER (English, Hindi)	en-Latn, hi-Deva	4,622 / 880 / 3,769
Stress language (binary)	Dreaddit (Reddit posts)	en-Latn	2,391 / 425 / 715

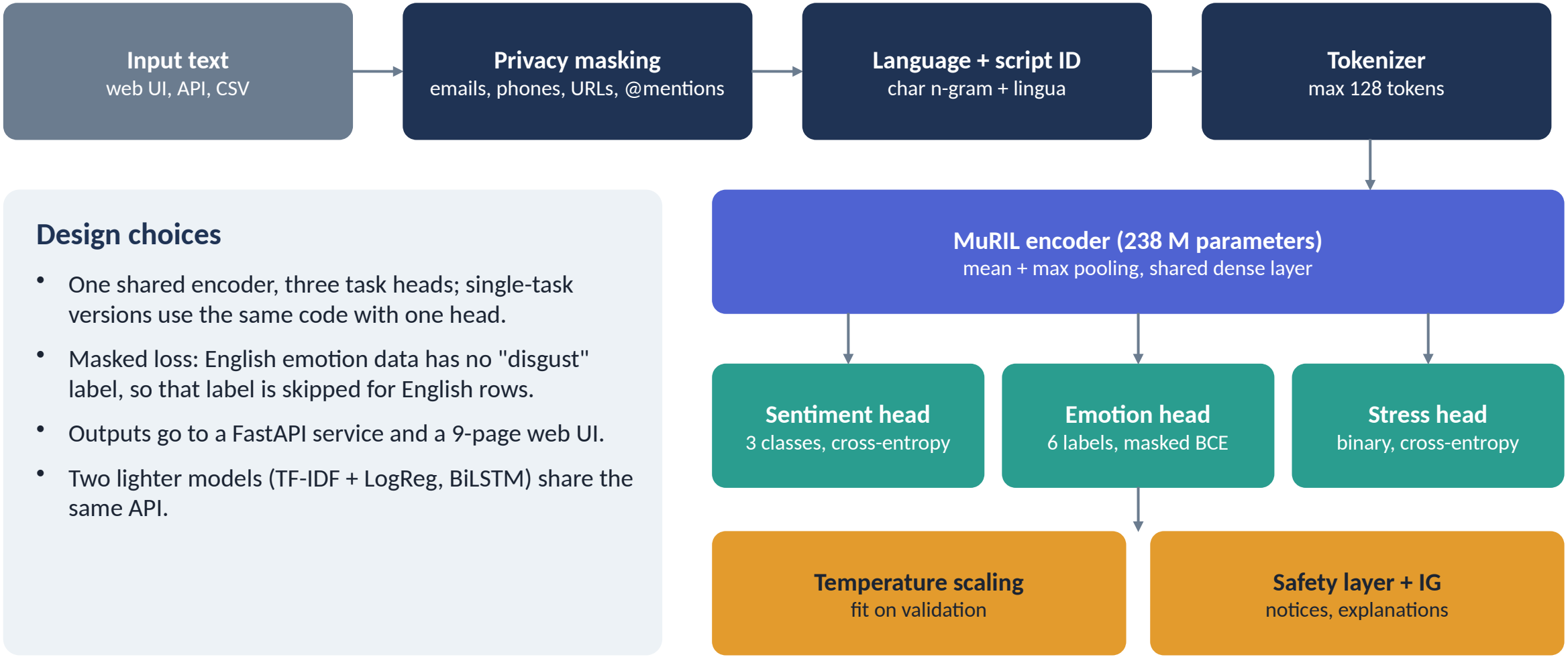
Each task comes from a different dataset and population, so results are reported per task and per language.

Integrity checks we added

- Frozen splits: SHA-1 hash per text and a SHA-256 manifest; a check script confirms anyone gets the same splits.
- Exact and near-duplicate removal across train, validation and test.
- Group splits by Reddit post and YouTube video, so one thread never spans two splits.
- Fixed a dedup bug that wrongly removed 4,002 Hindi texts (Devanagari vowel signs were being stripped).

System architecture

Rubric 1



Design choices

- One shared encoder, three task heads; single-task versions use the same code with one head.
- Masked loss: English emotion data has no "disgust" label, so that label is skipped for English rows.
- Outputs go to a FastAPI service and a 9-page web UI.
- Two lighter models (TF-IDF + LogReg, BiLSTM) share the same API.

Alternatives compared and validation

Rubric 1

Option	Why it was tried	Outcome
TF-IDF + logistic regression	Fast, interpretable baseline	Kept as baseline
Multi-task BiLSTM	Small neural model, no pre-training	Kept as baseline
Multi-task MuRIL	Indian-language pre-training	Served model in the app
Single-task MuRIL	Check whether sharing helps	Better stress ranking
HingRoBERTa (sentiment)	Pre-trained on Roman-script Hinglish	Best sentiment model
256 tokens, head+tail input	23.6% of stress posts exceed 128 tokens	Rejected on validation

Validation protocol

- 3 random seeds (13, 21, 42) per configuration; mean and s.d. reported.
- Model selection on validation only. The test set is used once, for comparisons fixed in advance.
- Paired bootstrap, 1,000 resamples, 95% confidence intervals.
- Calibration checked with ECE and Brier score.
- Original single run reproduced exactly with seed 13.

2

Implementation and live demo

What is built, and the working web application on real inputs

Implementation status

Rubric 2

- **Data pipeline**
  download, cleaning, dedup, frozen splits
- **Language ID**
  en / hi-Deva / hi-Latn classifier
- **4 model families**
  TF-IDF, BiLSTM, MuRIL, HingRoBERTa
- **Training engine**
  multi-seed, resumable every 10 min
- **Calibration**
  temperature scaling, tuned thresholds
- **Evaluation**
  bootstrap CIs, error analysis
- **Explanations**
  Integrated Gradients per token
- **Web app + API**
  FastAPI, 9 pages, batch CSV
- **Privacy + safety**
  no text logged, PII masking, notices
- **Tests**
  68 unit tests, all passing

21

training runs on CPU (7 configurations x 3 seeds)

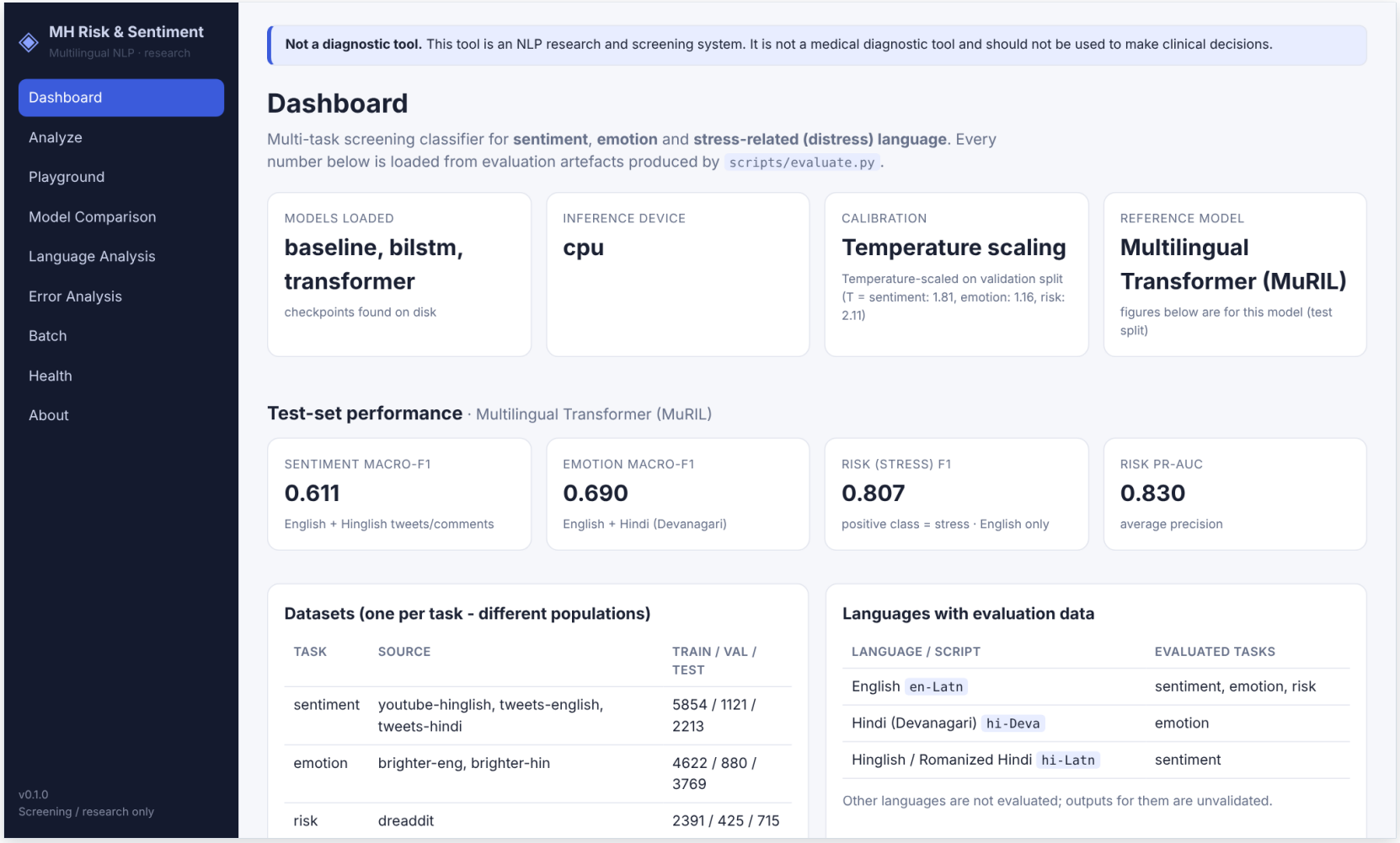
2,893 lines of Python in src/, 809 in scripts/, 713 in tests/

Not done yet

- Transliteration / spelling normalisation
- Learning-rate grid and XLM-R control (need a GPU)
- Manual error taxonomy (sheet ready)
- Sarvam adapter tested with mocks only

Live web app: dashboard

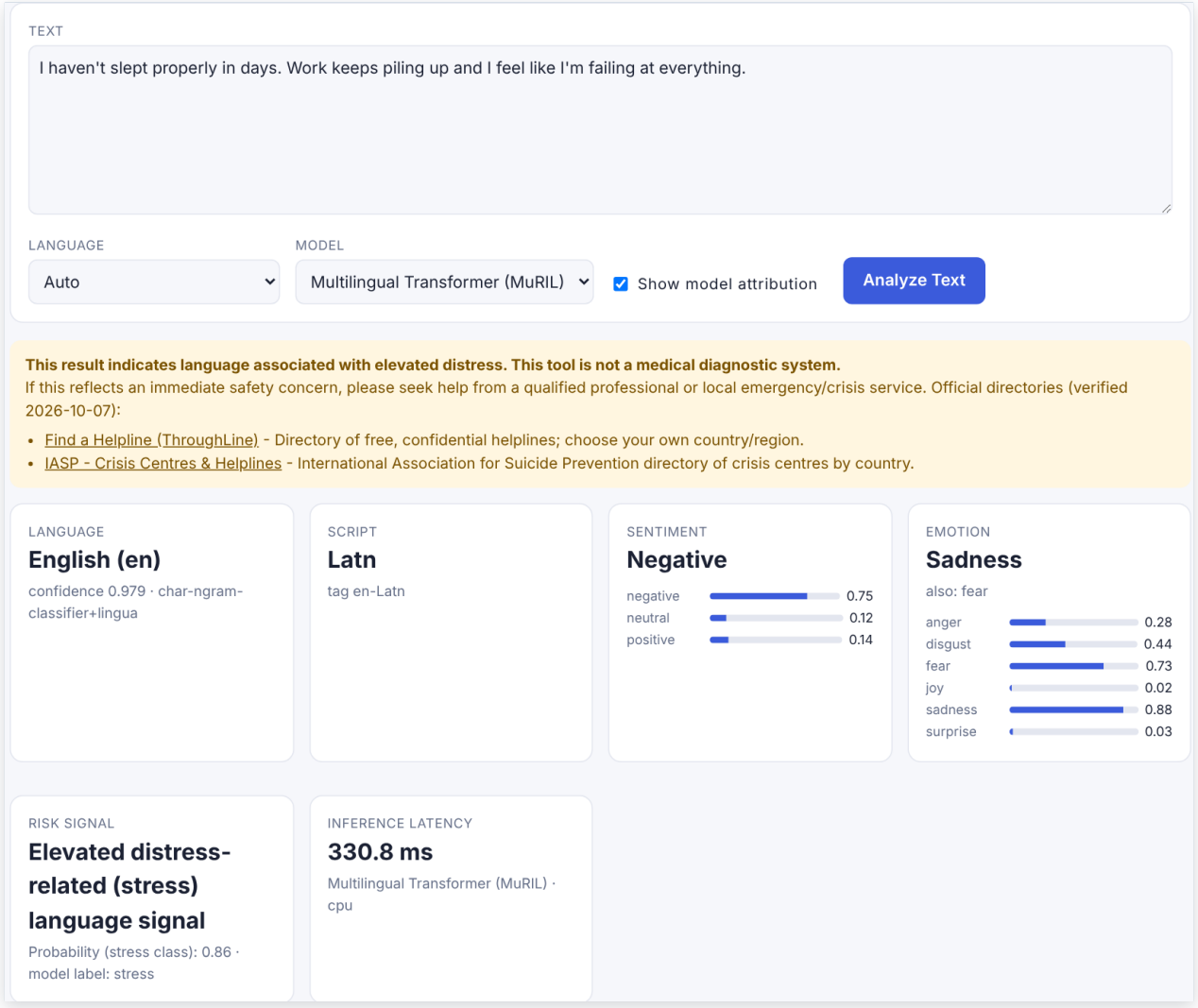
Rubric 2



- Shows which models loaded from disk (TF-IDF, BiLSTM, MuRIL) and the device.
- Test metrics are read from evaluation files, never typed in by hand.
- A banner on every page says the tool is not diagnostic.
- Run locally with one command: uvicorn src.app.main:app_factory --factory

Live example 1: English post

Rubric 2



Input: "I haven't slept properly in days. Work keeps piling up and I feel like I'm failing at everything."

Language: English, confidence 0.979

Sentiment: Negative (0.75)

Emotion: Sadness 0.88, fear 0.73

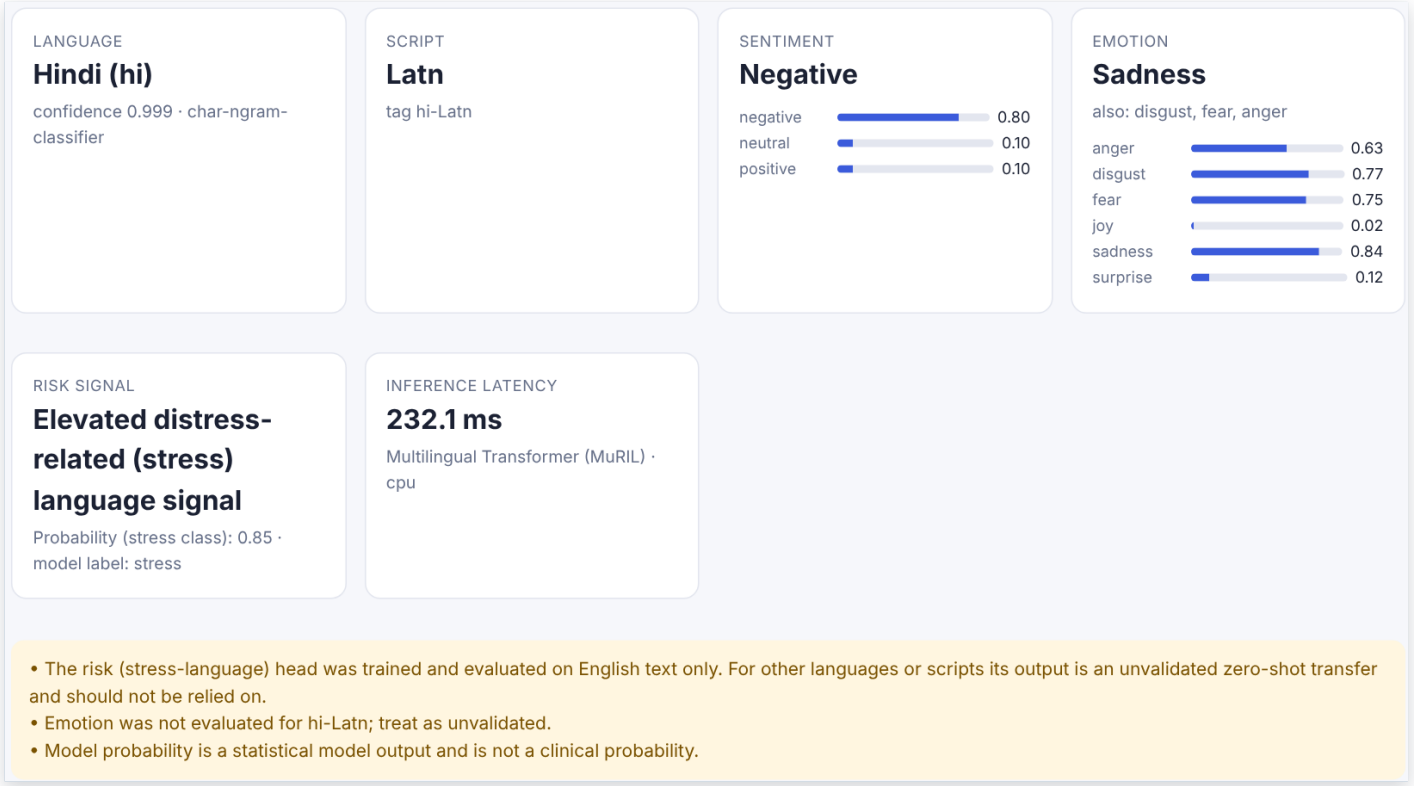
Stress signal: Elevated, probability 0.86

Response: Safety notice with official helpline directories

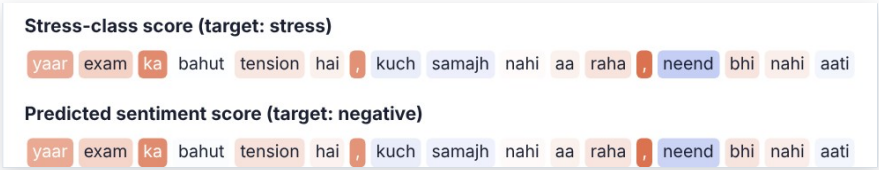
Live example 2: Hinglish post with explanation

Rubric 2

Input: "yaar exam ka bahut tension hai, kuch samajh nahi aa raha, neend bhi nahi aati"

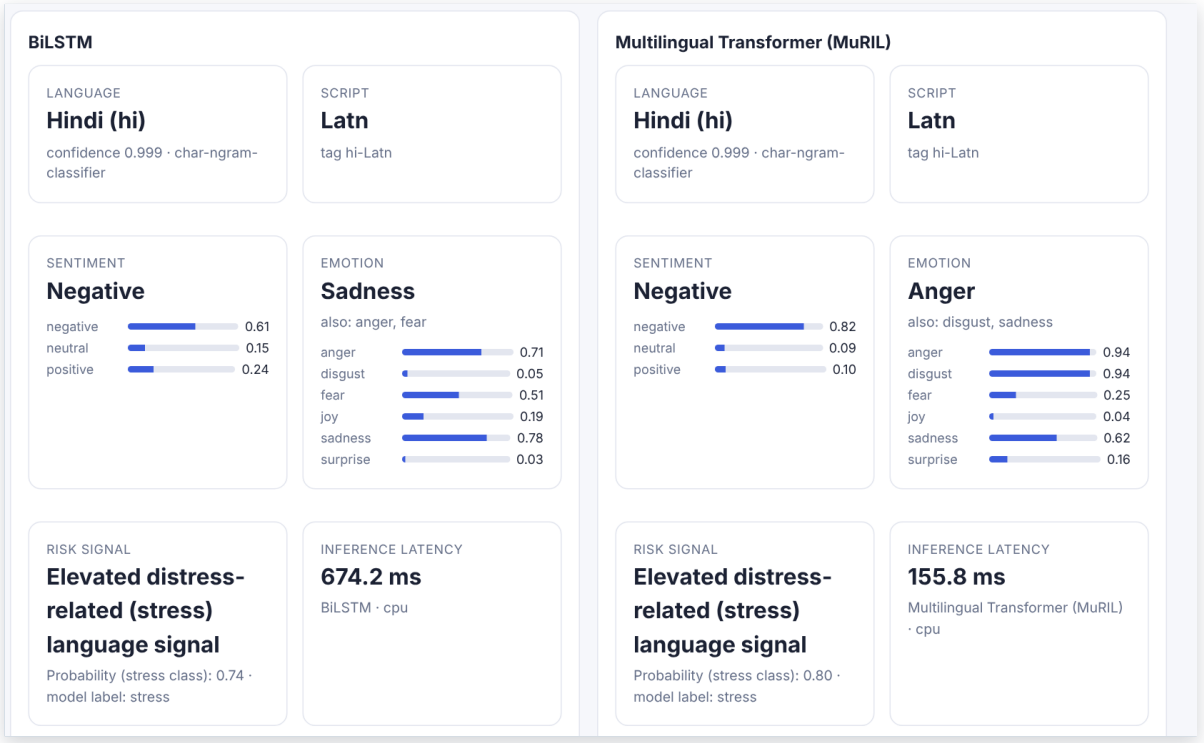


- Detected as Romanized Hindi (hi-Latn), confidence 0.999.
- The app warns that the stress head was trained on English only, so this output is unvalidated for Hinglish.
- Token attribution below: blue pushes toward the label, orange away. Our deletion test found it only weakly faithful for Hinglish (58% of examples).

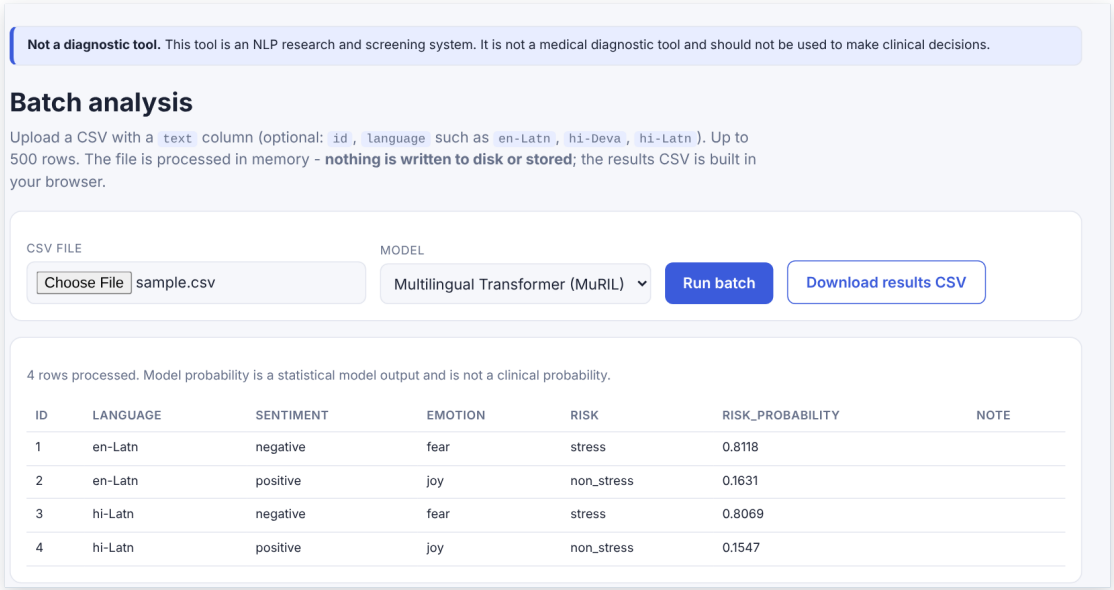


Model playground and batch analysis

Rubric 2



Same Hinglish sentence through BiLSTM and MuRIL side by side. Both flag stress; their emotion outputs differ (sadness vs anger).



A 4-row CSV (2 English, 2 Hinglish) processed in memory; nothing is written to disk. Up to 500 rows per file.

Anyone can clone and run it

Rubric 2

```
git clone https://github.com/faizdevx/
    Multilingual-Mental-Health-Risk-.git
pip install -r requirements.txt
python -m pytest -q          # 68 tests
python scripts/download_data.py
python scripts/prepare_data.py
python scripts/train_baseline.py
uvicorn src.app.main:app_factory --factory
```

Step-by-step guide in README.md: install, check, data, training, app, reproducing experiments.

Measured latency (median, 30 requests, 4-core CPU)

10.5 ms	TF-IDF + LogReg
11.7 ms	BiLSTM
124.5 ms	MuRIL transformer

End-to-end HTTP time for all three heads, without explanations.

3

Results and validation

Test-set results, mean of 3 seeds, with paired bootstrap 95% confidence intervals

Headline test results

Rubric 3

Model	Sentiment macro-F1	Emotion macro-F1	Stress F1	Stress PR-AUC	Stress ECE
Multi-task MuRIL (baseline)	0.603 ± 0.008	0.684 ± 0.011	0.796 ± 0.014	0.852 ± 0.029	0.035 ± 0.002
Single-task MuRIL	0.606 ± 0.003	0.682 ± 0.002	0.799 ± 0.005	**0.875 ± 0.010**	0.062 ± 0.023
Single-task HingRoBERTa	**0.648 ± 0.013**	-	-	-	-

+0.056

Hinglish sentiment macro-F1 from HingRoBERTa, 95% CI [+0.032, +0.079]

+0.023

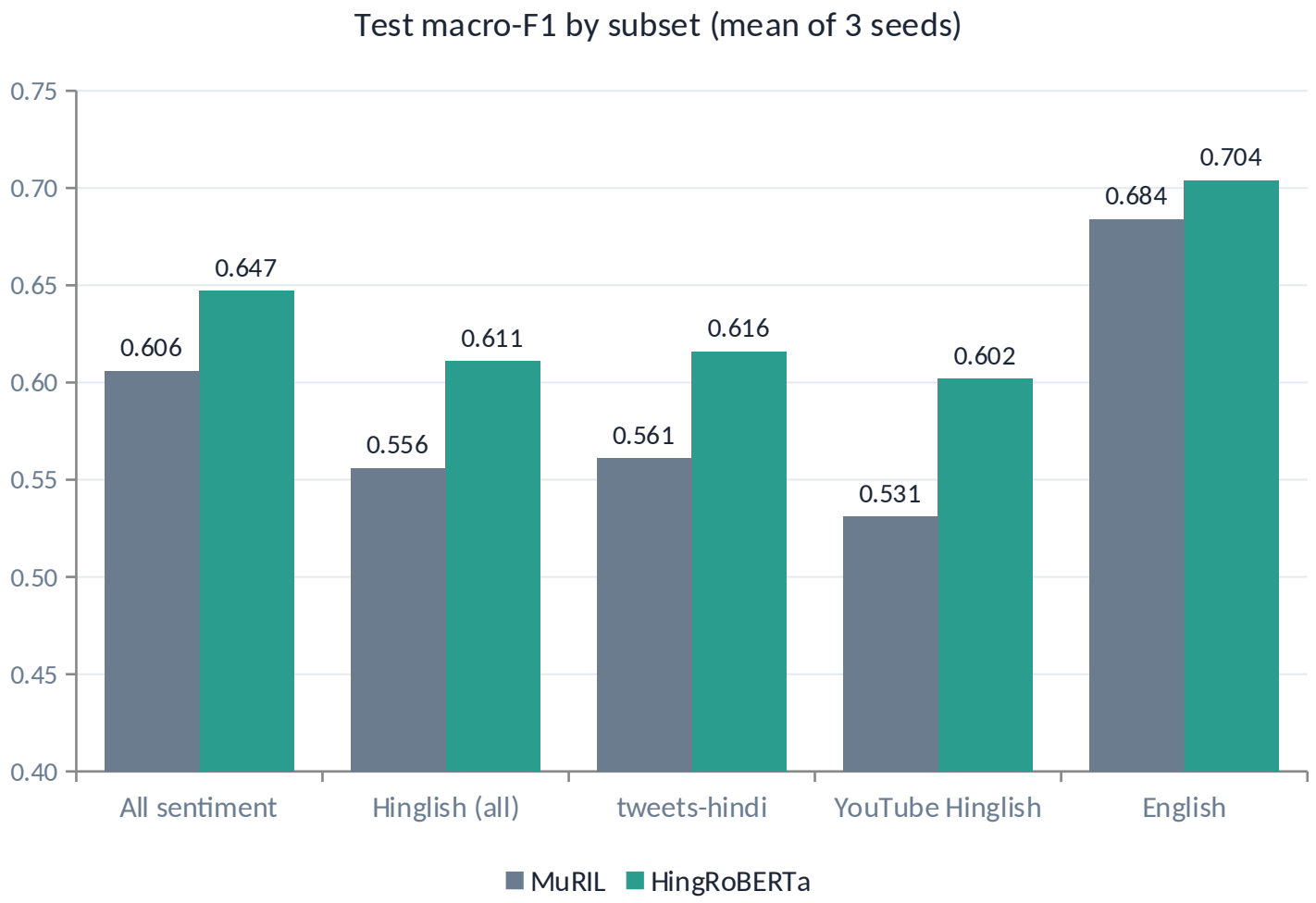
stress PR-AUC from training stress alone, CI [+0.008, +0.038]

0

tasks that gained from multi-task sharing

Hinglish pre-training narrows the gap

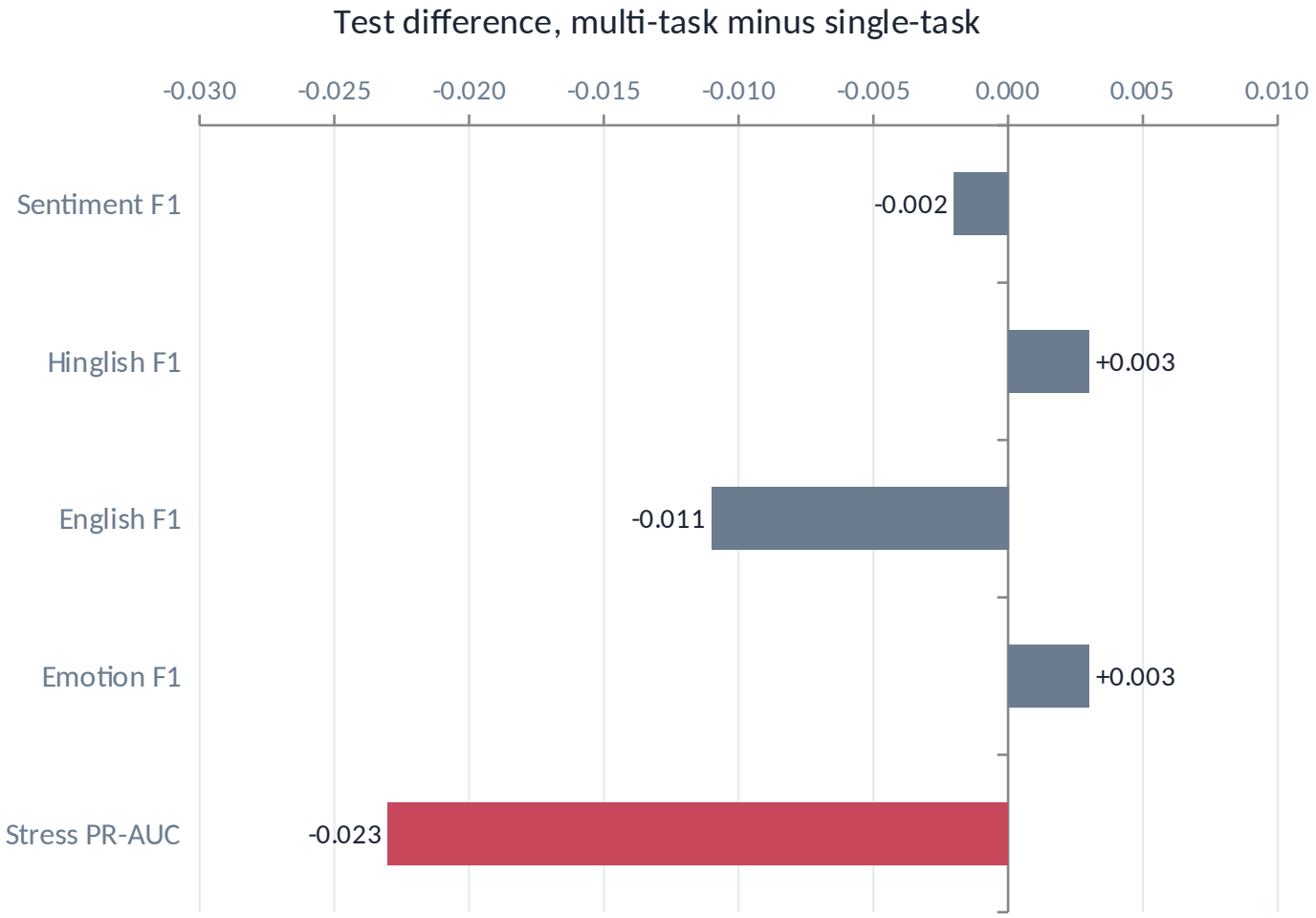
Rubric 3



- Hinglish: 0.556 to 0.611, +0.056 [+0.032, +0.079].
- Gains on both Hinglish sources: tweets +0.054, YouTube +0.071.
- English unchanged: +0.020 [-0.002, +0.044].
- English to Hinglish gap: 0.128 to 0.093.
- Contamination check: 0 test texts found in HingRoBERTa's 52.9 M-line pre-training corpus.

Multi-task sharing did not help

Rubric 3



Sharing

Only stress PR-AUC changed significantly: -0.023 [-0.038, -0.008]. The three datasets come from different platforms and annotators, so the tasks interfere.

Negative result: longer input

Head+tail 256 tokens lowered validation macro-F1 by 0.020 [-0.037, -0.003]. Plain 256 tokens made no difference. Neither went to test.

Reliability: abstention and explanations

Rubric 3

Abstain on low-confidence inputs

0.783 to 0.828

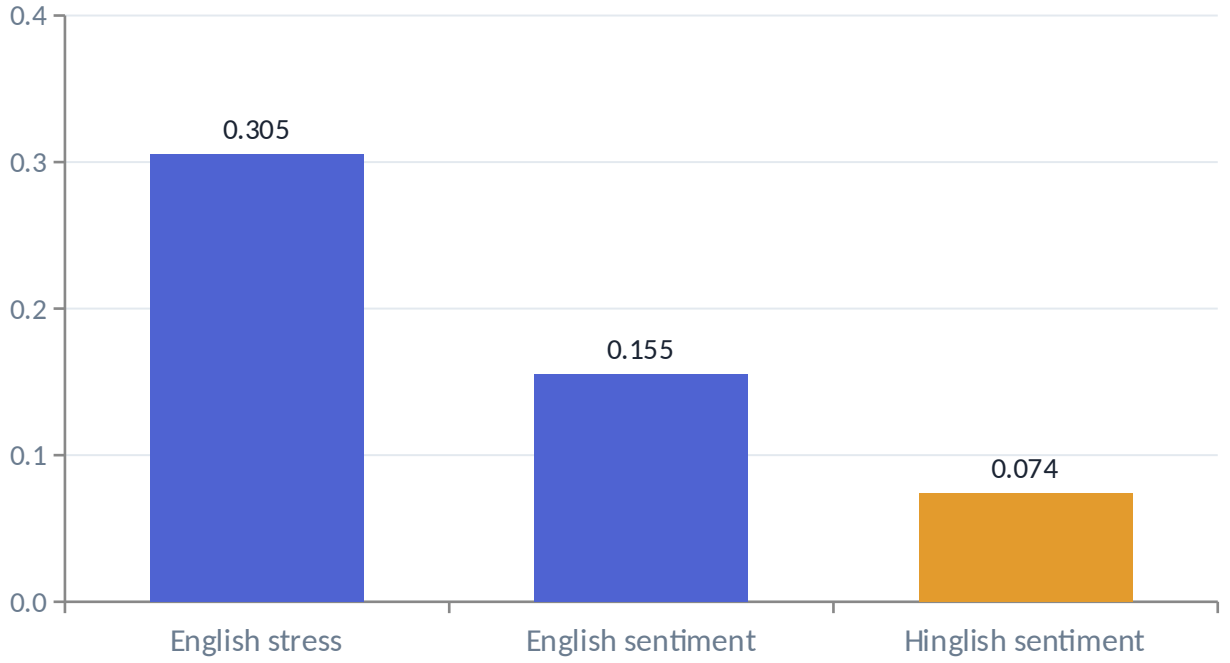
stress macro-F1 on the 83% of inputs the model answers

0.647 to 0.702

HingRoBERTa sentiment macro-F1 at 74% coverage

Cut-off chosen on validation (20th percentile of confidence) and applied unchanged to test. Low confidence describes the text, not the person.

IG deletion test: extra probability drop vs random tokens



Explanations are faithful for English stress (92% of examples) but weak for Hinglish (58%). The app's explanation note says so.

Comparison with published research

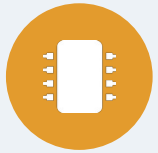
Rubric 3

Task and dataset	Published (source)	Ours (test, 3 seeds)	Comparable?	Reading
Stress, Dreaddit	BERT-base F1 0.8065 (Turcan & McKeown, 2019)	F1 0.798 (single-task MuRIL)	Yes, same 715-post test set	On par
Hinglish sentiment	HingRoBERTa +4.86 over XLM-R (Nayak & Joshi, 2022)	+5.6 over MuRIL	Different data	Effect replicated
Hindi tweets, UMSAB	XLM-Twitter 56.39 (Barbieri et al., 2022)	HingRoBERTa 0.616	Same test file, different training mix	About 5 points above
English tweets, UMSAB	XLM-Twitter 70.63	HingRoBERTa 0.704	Partly	On par
Emotion, BRIGHTER English	RoBERTa baseline 0.708	0.639 to 0.645	Partly	Below
Emotion, BRIGHTER Hindi	RoBERTa baseline 0.855	0.768 to 0.770	Partly	Below

Emotion is our weakest task: 3 epochs on CPU and less training data, since validation was carved from train.

Limitations we report

Rubric 3



Compute

CPU only: 3 seeds, 3 epochs, no learning-rate search. Best epoch was often the last, so models may be under-trained.



Language coverage

Stress labels exist for English only. Hindi in Devanagari has emotion labels only; Hinglish has sentiment only.



Data quality

Hinglish data is small and noisy. The tweets-hindi validation split scores near chance for every model; cause unknown.



Clinical meaning

Stress is a crowd-annotated label on Reddit posts. It is not a clinical status and was not validated on other populations.

4

Team, planning and scalability

Who did what, how resources were used, and how the system scales

Team work and individual contribution

Rubric 5

Faisal Arif Ansari	Paras Tiwari	Prateek Rawat	Lakshay Mishra

Work package	Main output	Owner(s)
Literature review and dataset survey	Related work, 37 verified references	
Data pipeline and split integrity	Download, cleaning, dedup, frozen splits	
Baselines (TF-IDF, BiLSTM) and language ID	3 baseline models, langid classifier	
Transformer experiments	21 runs: MuRIL, HingRoBERTa, ablations	
Evaluation and statistics	Bootstrap CIs, calibration, faithfulness	
Web app, API, privacy and safety	FastAPI service, 9-page UI, tests	
Research paper and presentation	IEEE paper, README, this deck	

Timeline, resources and cost

Rubric 6

7 Oct
System built: data pipeline, 3 models, web app

22 Oct
Research audit, experiment plan, frozen splits

22-24 Oct
21 CPU training runs with resumable snapshots

24 Oct
Results, conclusion, IEEE paper

26 Oct
Clone-and-run guide for anyone

Resource	What we used	Cash cost
Compute	One 4-core CPU, 16 GB RAM; at least 23.3 h of recorded training time for 21 runs	Rs 0
GPU	None. Kaggle GPU requests never attached, so training was made resumable instead	Rs 0
Software	PyTorch, Transformers, scikit-learn, FastAPI (open source)	Rs 0
Data	4 public datasets; no new data collected from people	Rs 0
Storage	940 MB of checkpoints (MuRIL 914 MB, others 13 MB each)	Rs 0

Scalability

Rubric 6

~8 texts/s

MuRIL, one process on 4 CPU cores (from 124.5 ms median)

~85 texts/s

TF-IDF or BiLSTM on the same machine (10.5-11.7 ms)

500 rows

per batch CSV upload, processed in memory

How it scales

- The API keeps no per-user state, so more replicas behind a load balancer add throughput linearly. Servers needed is roughly peak texts per second divided by 8.
- High-volume, low-stakes traffic can go to the 13 MB TF-IDF model, about 12 times faster than MuRIL.
- New languages need labelled data per task; the code is language-agnostic, the claims are not.
- Planned, not yet measured: GPU serving, batched inference, 8-bit quantisation.

5

Ethics and research paper

Responsible design, and the state of the paper

Ethics, privacy, safety and sustainability

Rubric 7

Not diagnostic

Disclaimers on every page and response. "Risk" is defined as stress-related language only.

Privacy

Submitted text is never logged or stored. Logs keep a whitelist of metadata. Emails, phones, URLs and @mentions are masked.

Safety response

Elevated signals show official helpline directories (checked 7 Oct 2026). No hard-coded phone numbers that could go stale.

Fairness

Metrics reported per language. Outputs for languages without evaluation data are labelled unvalidated.

Data ethics

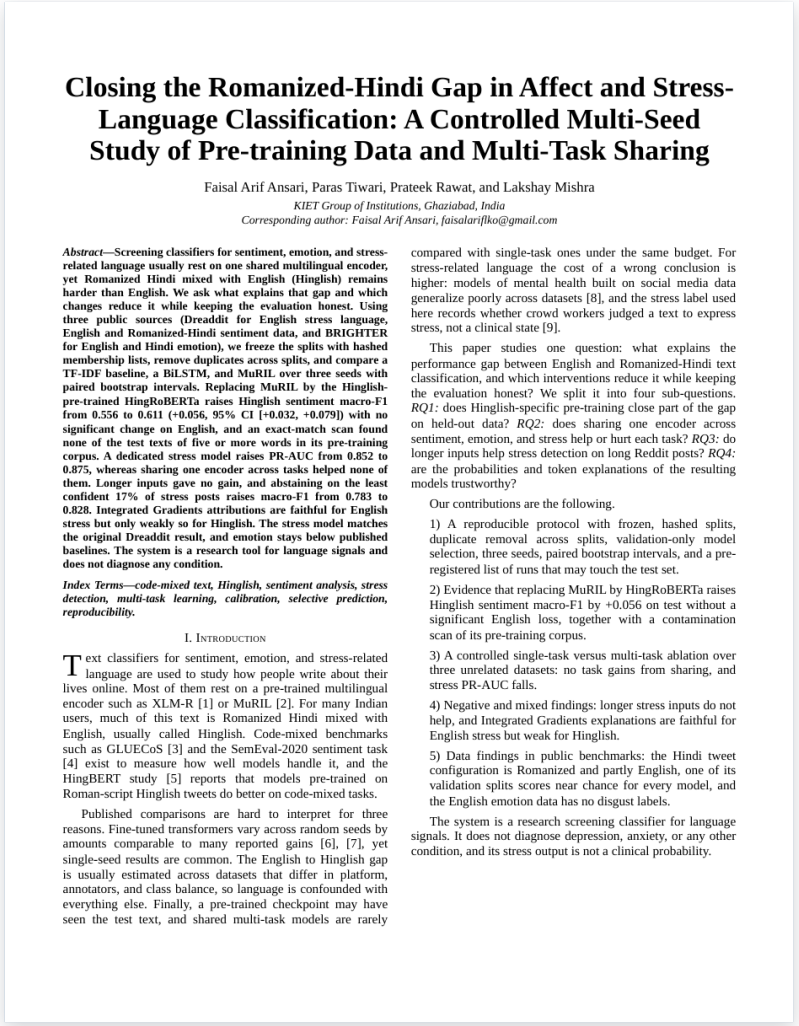
Public datasets only; raw Reddit and Twitter text is not redistributed. Dreaddit's licence is unverified and flagged.

Environment

CPU only, no GPU hours. Reused pre-trained encoders. 13 MB models available for low-energy serving.

Research paper and submission readiness

Rubric 8, 9



Closing the Romanized-Hindi Gap in Affect and Stress-Language Classification: A Controlled Multi-Seed Study of Pre-training Data and Multi-Task Sharing

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Abstract—Screening classifiers for sentiment, emotion, and stress-related language usually rest on one shared multilingual encoder, yet Romanized Hindi mixed with English (Hinglish) remains harder than English. We ask what explains that gap and which changes reduce it while keeping the evaluation honest. Using three public sources (Dreaddit for English stress language, English and Romanized-Hindi sentiment data, and BRIGHTER for English and Hindi emotion), we freeze the splits with hashed membership lists, remove duplicates across splits, and compare a TF-IDF baseline, a BiLSTM, and MuRIL over three seeds with paired bootstrap intervals. Replacing MuRIL by the Hinglish-pre-trained HingRoBERTa raises Hinglish sentiment macro-F1 from 0.556 to 0.611 (+0.056, 95% CI [+0.032, +0.079]) with no significant change on English, and an exact-match scan found none of the test texts of five or more words in its pre-training corpus. A dedicated stress model raises PR-AUC from 0.852 to 0.875, whereas sharing one encoder across tasks helped none of them. Longer inputs gave no gain, and abstaining on the least confident 17% of stress posts raises macro-F1 from 0.783 to 0.828. Integrated Gradients attributions are faithful for English stress but only weakly so for Hinglish. The stress model matches the original Dreaddit result, and emotion stays below published baselines. The system is a research tool for language signals and does not diagnose any condition.

***Index Terms*—code-mixed text, Hinglish, sentiment analysis, stress detection, multi-task learning, calibration, selective prediction, reproducibility.**

I. Introduction

Text classifiers for sentiment, emotion, and stress-related language are used to study how people write about their lives online. Most of them rest on a pre-trained multilingual encoder such as XLM-R [1] or MuRIL [2]. For many Indian users, much of this text is Romanized Hindi mixed with English, usually called Hinglish. Code-mixed benchmarks such as GLUECoS [3] and the SemEval-2020 sentiment task [4] exist to measure how well models handle it, and the HingBERT study [5] reports that models pre-trained on Roman-script Hinglish tweets do better on code-mixed tasks.

Published comparisons are hard to interpret for three reasons. Fine-tuned transformers vary across random seeds by amounts comparable to many reported gains [6], [7], yet single-seed results are common. The English to Hinglish gap is usually estimated across datasets that differ in platform, annotators, and class balance, so language is confounded with everything else. Finally, a pre-trained checkpoint may have seen the test text, and shared multi-task models are rarely compared with single-task ones under the same budget. For stress-related language the cost of a wrong conclusion is higher: models of mental health built on social media data generalize poorly across datasets [8], and the stress label used here records whether crowd workers judged a text to express stress, not a clinical state [9].

This paper studies one question: what explains the performance gap between English and Romanized-Hindi text classification, and which interventions reduce it while keeping the evaluation honest? We split it into four sub-questions. *RQ1:* does Hinglish-specific pre-training close part of the gap on held-out data? *RQ2:* does sharing one encoder across sentiment, emotion, and stress help or hurt each task? *RQ3:* do longer inputs help stress detection on long Reddit posts? *RQ4:* are the probabilities and token explanations of the resulting models trustworthy?

Our contributions are the following.

1) A reproducible protocol with frozen, hashed splits, duplicate removal across splits, validation-only model selection, three seeds, paired bootstrap intervals, and a pre-registered list of runs that may touch the test set.

2) Evidence that replacing MuRIL by HingRoBERTa raises Hinglish sentiment macro-F1 by +0.056 on test without a significant English loss, together with a contamination scan of its pre-training corpus.

3) A controlled single-task versus multi-task ablation over three unrelated datasets: no task gains from sharing, and stress PR-AUC falls.

4) Negative and mixed findings: longer stress inputs do not help, and Integrated Gradients explanations are faithful for English stress but weak for Hinglish.

5) Data findings in public benchmarks: the Hindi tweet configuration is Romanized and partly English, one of its validation splits scores near chance for every model, and the English emotion data has no disgust labels.

The system is a research screening classifier for language signals. It does not diagnose depression, anxiety, or any other condition, and its stress output is not a clinical probability.

Closing the Romanized-Hindi Gap in Affect and Stress-Language Classification: A Controlled Multi-Seed Study of Pre-training Data and Multi-Task Sharing

- IEEE conference format, 7 pages, 8 tables, 4 figures.
- 37 references, each checked against arXiv, ACL Anthology or Crossref.
- Every number traced to a run file in the public repository.
- Contributions: Hinglish pre-training gain, multi-task ablation, truncation negative result, language-dependent explanation faithfulness, two data-quality findings.

Done Paper PDF and source

Done Public code, frozen splits, run logs

Done Reproducibility guide in README

To do Plagiarism report (institute tool)

To do Target venue chosen

Conclusion and next steps

What we showed

- Hinglish pre-training improved Hinglish sentiment by +0.056 and closed 27% of the gap.
- Multi-task sharing gave no gain and hurt stress ranking; a separate stress model raised PR-AUC to 0.875.
- Our stress model matches the original Dreaddit result (F1 0.798 vs 0.8065).
- A working, privacy-preserving web app that anyone can clone and run.

Next steps, in order

1. Train emotion longer; try RemBERT
2. Learning-rate grid and 5 seeds on a GPU
3. Hand-check the tweets-hindi validation labels
4. Finish the manual Hinglish error taxonomy
5. Test transliteration to Devanagari
6. Serve HingRoBERTa for Hinglish in the app

Thank you

Questions and live demo

github.com/faizdevx/Multilingual-Mental-Health-Risk-

Faisal Arif Ansari | Paras Tiwari | Prateek Rawat | Lakshay Mishra

KIET Group of Institutions, Ghaziabad